

Cloud_Computing_Practical_Exam

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EXCS A

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Q1:

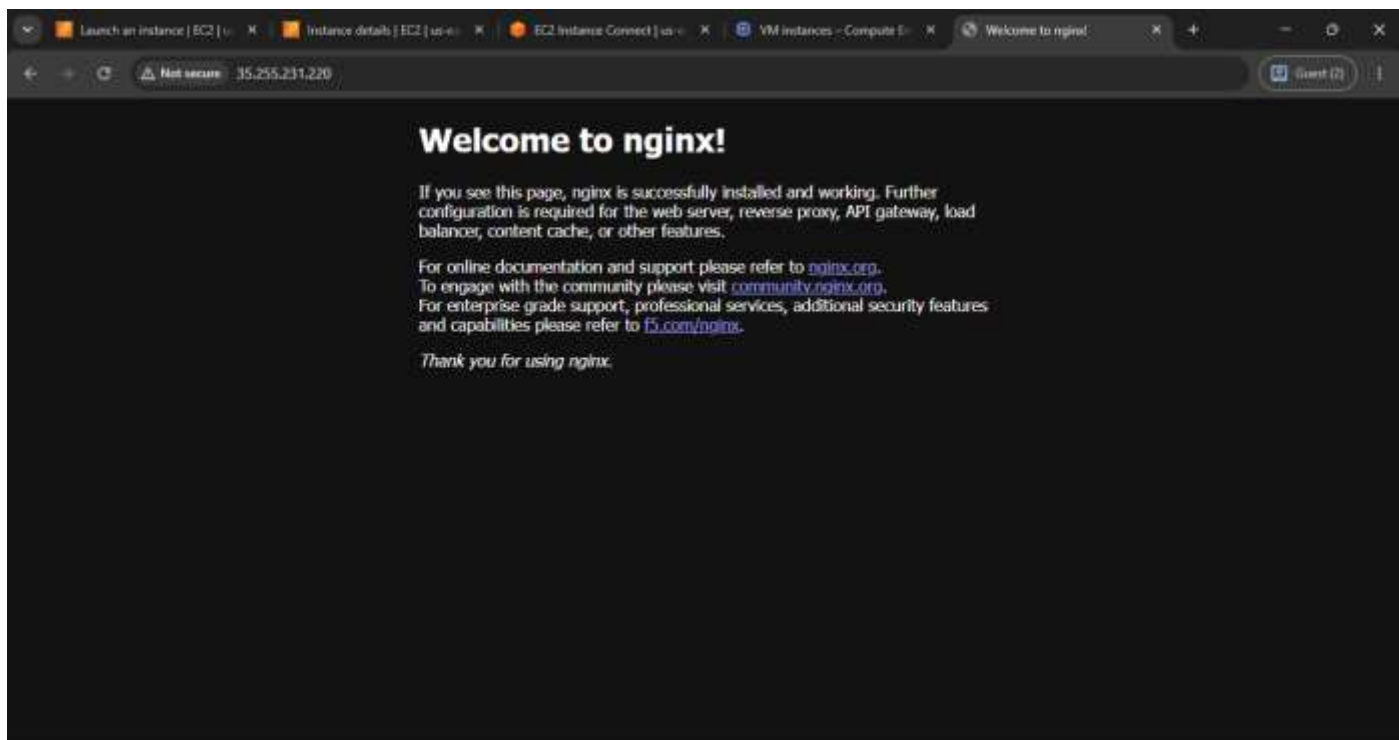
```
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari?authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari?authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true
SSH-in-browser
UPLOAD FILE DOWNLOAD FILE

Setting up libnftables1:amd64 (1.0.6-2+deb12u2) ...
Setting up docker.io (20.10.24+dfsg1-1+deb12u1+b6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /lib/systemd/system/docker.socket.
Setting up perl (5.36.0-7+deb12u3) ...
Setting up libgprofng0:amd64 (2.40-2) ...
Setting up libmodule-find-perl (0.16-2) ...
Setting up libproc-processtable-perl:amd64 (0.634-1+b2) ...
Setting up criu (3.17.1-2+deb12u2) ...
Setting up libintl-perl (1.33-1) ...
Setting up libterm-readkey-perl (2.38-2+b1) ...
Setting up libsort-naturally-perl (1.03-4) ...
Setting up binutils-x86-64-linux-gnu (2.40-2) ...
Setting up binutils (2.40-2) ...
Setting up libintl-xs-perl (1.33-1) ...
Setting up liberror-perl (0.17029-2) ...
Setting up git (1:2.39.5-0+deb12u3) ...
Setting up needrestart (3.6-4+deb12u3) ...
Processing triggers for man-db (2.11.2-2) ...
Processing triggers for libc-bin (2.36-9+deb12u14) ...
Synchronizing state of docker.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable docker
aribahshirgaonker@abdulmannan-ansari:~$ docker --version
Docker version 20.10.24+dfsg1, build 297a128
aribahshirgaonker@abdulmannan-ansari:~$
```

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ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari?authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari?authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true
SSH-in-browser
UPLOAD FILE DOWNLOAD FILE

aribahshirgaonker@abdulmannan-ansari:~$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
aribahshirgaonker@abdulmannan-ansari:~$
```

```
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser=0&file=en_GB&projectNumber=814985417380&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser=0&file=en_GB&projectNumber=814985417380&useAdminProxy=true
SSH in-browser
UPLOAD FILE
DOWNLOAD FILE
aribahshirgaonker@abdulmannan-ansari:~$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
aribahshirgaonker@abdulmannan-ansari:~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
72c03230f136: Pull complete
d4dcde3aeed: Pull complete
c5a7565de4cf: Pull complete
57b3fbf43092: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
6376498be516: Pull complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
aribahshirgaonker@abdulmannan-ansari:~$ docker run -d --name my-nginx -p 80:80 nginx
7f827815fec4dd209900c861de4a5e01925b7f8b7b82e840162dc04d350ee63
aribahshirgaonker@abdulmannan-ansari:~$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
7f827815fec4 nginx "/docker-entrypoint..." 6 seconds ago Up 5 seconds 0.0.0.0:80->80/tcp, :::80->80/tcp my-nginx
aribahshirgaonker@abdulmannan-ansari:~$
```



Q2:

```
ssh.cloud.google.com/v2/ssh/projects/vit-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/vit-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true
SSH in-browser
aribahshirgaonker@abdulmannan-ansari:~$ docker volume create studentdata
studentdata
aribahshirgaonker@abdulmannan-ansari:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
aribahshirgaonker@abdulmannan-ansari:~$ docker run -it --name ubuntu-container -v studentdata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335ddd46: Pull complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b0b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@a2f6d081ca71:/#
```

```
ssh.cloud.google.com/v2/ssh/projects/vit-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/vit-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true
SSH in-browser
aribahshirgaonker@abdulmannan-ansari:~/my-docker-app$ docker run -it --name ubuntu-container -v studentdata:/data ubuntu bash
root@a2f6d081ca71:/# mkdir -p /data
echo "Name: Abdulmannan Ansari | Roll No: 24108A0022" > /data/info.txt

cat /data/info.txt
Name: Abdulmannan Ansari | Roll No: 24108A0022
root@a2f6d081ca71:/#
```

Q3:

```
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser-0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser-0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true
SSH in-browser
GNU nano 7.2 Dockerfile
FROM alpine:latest
CMD ["echo", "Cloud Computing Practical Exam Completed Successfully"]
Help      Write Out  Where Is   Cut        Read 3 lines  Execute    Location    Undo      Set Mark
Exit      Read File  Replace    Paste      Justify        Go To Line  Redo      Copy
```

```
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser-0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser-0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true
SSH in-browser
aribahshirgaonker@abdulmannan-ansari:~$ mkdir my-docker-app && cd my-docker-app
nano Dockerfile
aribahshirgaonker@abdulmannan-ansari:~/my-docker-app$ nano Dockerfile
aribahshirgaonker@abdulmannan-ansari:~/my-docker-app$ docker build -t practical-exam-image .
Sending build context to Docker daemon  2.048kB
Step 1/2 : FROM alpine:latest
latest: Pulling from library/alpine
5b76ea313681f: Pull complete
Digest: sha256:a2d49ea696c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c029f4
Status: Downloaded newer image for alpine:latest
--> 772baa4540c4
Step 2/2 : CMD ["echo", "Cloud Computing Practical Exam Completed Successfully"]
--> Running in e9e6ea260fd2
Removing intermediate container e9e6ea260fd2
--> 7f045cd97a43
Successfully built 7f045cd97a43
Successfully tagged practical-exam-image:latest
aribahshirgaonker@abdulmannan-ansari:~/my-docker-app$
```

```
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true
SSH in-browser
UPLOAD FILE
DOWNLOAD FILE
arihahshirgaonker@abdulmannan-ansari:~$ mkdir my-docker-app && cd my-docker-app
nano Dockerfile
arihahshirgaonker@abdulmannan-ansari:~/my-docker-app$ nano Dockerfile
arihahshirgaonker@abdulmannan-ansari:~/my-docker-app$ docker build -t practical-exam-image .
Sending build context to Docker daemon 2.048kB
Step 1/2 : FROM alpine:latest
latest: Pulling from library/alpine
5b70e313681f: Pull complete
Digest: sha256:a2d49ea696c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c029f4
Status: Downloaded newer image for alpine:latest
--> 772bee4540c4
Step 2/2 : CMD ["echo", "Cloud Computing Practical Exam Completed Successfully"]
--> Running in e9e6ea260fd2
Removing intermediate container e9e6ea260fd2
--> 7f045cd97a43
Successfully built 7f045cd97a43
Successfully tagged practical-exam-image:latest
arihahshirgaonker@abdulmannan-ansari:~/my-docker-app$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
practical-exam-image latest          7f045cd97a43    About a minute ago  8.45MB
nginx               latest          936fef290c8f    2 days ago      161MB
alpine              latest          772bee4540c4    3 days ago      8.45MB
ubuntu              latest          30ba44506a6d    7 weeks ago     100MB
arihahshirgaonker@abdulmannan-ansari:~/my-docker-app$
```

```
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/vt-vita-abdulmannan/zones/us-central1-c/instances/abdulmannan-ansari/authuser=0&hl=en_GB&projectNumber=814985417380&useAdminProxy=true
SSH in-browser
UPLOAD FILE
DOWNLOAD FILE
arihahshirgaonker@abdulmannan-ansari:~$ mkdir my-docker-app && cd my-docker-app
nano Dockerfile
arihahshirgaonker@abdulmannan-ansari:~/my-docker-app$ nano Dockerfile
arihahshirgaonker@abdulmannan-ansari:~/my-docker-app$ docker build -t practical-exam-image .
Sending build context to Docker daemon 2.048kB
Step 1/2 : FROM alpine:latest
latest: Pulling from library/alpine
5b70e313681f: Pull complete
Digest: sha256:a2d49ea696c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c029f4
Status: Downloaded newer image for alpine:latest
--> 772bee4540c4
Step 2/2 : CMD ["echo", "Cloud Computing Practical Exam Completed Successfully"]
--> Running in e9e6ea260fd2
Removing intermediate container e9e6ea260fd2
--> 7f045cd97a43
Successfully built 7f045cd97a43
Successfully tagged practical-exam-image:latest
arihahshirgaonker@abdulmannan-ansari:~/my-docker-app$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
practical-exam-image latest          7f045cd97a43    About a minute ago  8.45MB
nginx               latest          936fef290c8f    2 days ago      161MB
alpine              latest          772bee4540c4    3 days ago      8.45MB
ubuntu              latest          30ba44506a6d    7 weeks ago     100MB
arihahshirgaonker@abdulmannan-ansari:~/my-docker-app$ docker run --rm practical-exam-image
Cloud Computing Practical Exam Completed Successfully
arihahshirgaonker@abdulmannan-ansari:~/my-docker-app$
```

Bonus Questions:

1. Docker Image vs. Docker Container

A Docker Image is a read-only, static template containing the application code, runtime, libraries, and environment configuration needed to run an application. A Docker Container is a runtime, isolated instance of that image that is active, writeable, and executes the actual application processes.

2. Why Docker Volumes Are Used

Docker Volumes are used to persistently store and manage data generated by running containers outside of the container's lifecycle. Since container file systems are temporary and destroyed when a container deletes, volumes ensure that critical data (like databases or logs) is safely preserved and can be easily shared between multiple containers.